

Sequential Learning

Home assignment

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1 Rock paper, scissors

1.1

Recall M , N and a loss matrix $L \in [-1, 1]^{M \times N}$ that corresponds to the game “Rock paper scissors”

The loss matrix for the Rock, paper, scissors game is

$$L = \begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix} \quad (1)$$

$M = N = 3$

Full information feedback

1.2 Simulation against a fixed adversary

Consider the game “Rock paper scissors” and assume that the adversary chooses $q_t = (1/3, 1/6, 1/2)$ and samples $j_t \sim q_t$ for all rounds $t \geq 1$

- a) What is the loss $l_t(i)$ incurred by the player if he chooses action i at time t ?

The loss $l_t(i)$ incurred by playing the action i at time t is: $L(i, j_t)$

- b) Simulate an instance of the game for $t = 1, \dots, T = 100$ for $\eta = 1$. Plot the evolution of the weight vectors $p_1, p_2, \dots, p_T$. What seems to be the best strategy against this adversary?

The best strategy against this fixed adversary seems to be playing always the action that would win against its favorite play. In this case, the agent learns to always play action 1 (rock) because the most probable action of the fixed opponent is action 3 (scissors).

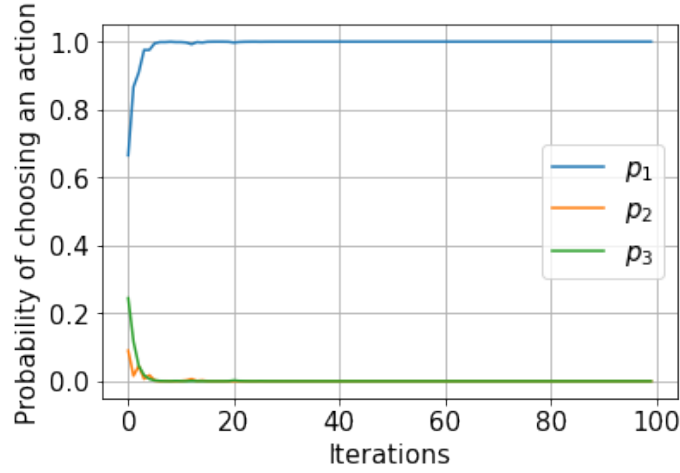


Figure 1: Evolution of the weight vectors p_i for $\eta = 1$

c) Plot the average loss $\hat{l}_t = \frac{1}{t} \sum_{s=1}^t l(i_s, j_s)$ as a function of t

The average loss ends up being negative, which shows that the agent is indeed learning a winning strategy. I.e in average is defeating its adversary. It converges to ≈ -0.3 and in this case the expected loss is $-\frac{1}{2} + \frac{1}{6} = -\frac{1}{3}$.

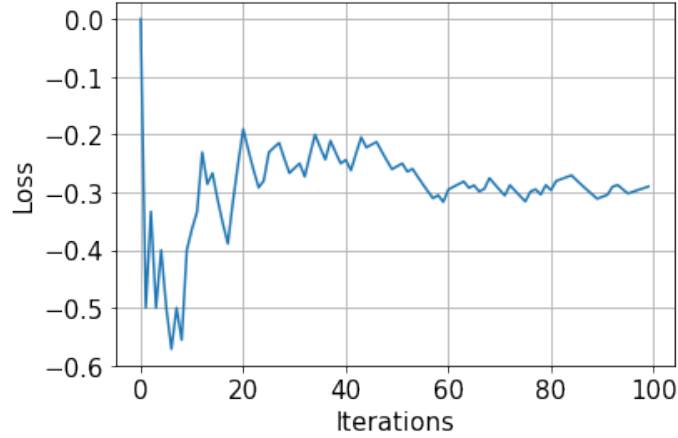


Figure 2: Evolution of average loss during 100 steps for $\eta = 1$

d) Repeat one simulation for different values of learning rates $\eta \in 0.01, 0.05, 0.1, 0.5, 1$. What are the best η in practice and in theory?

When running the experiments for $\eta \in 0.01, 0.05, 0.1, 0.5, 1$ we observe the following results

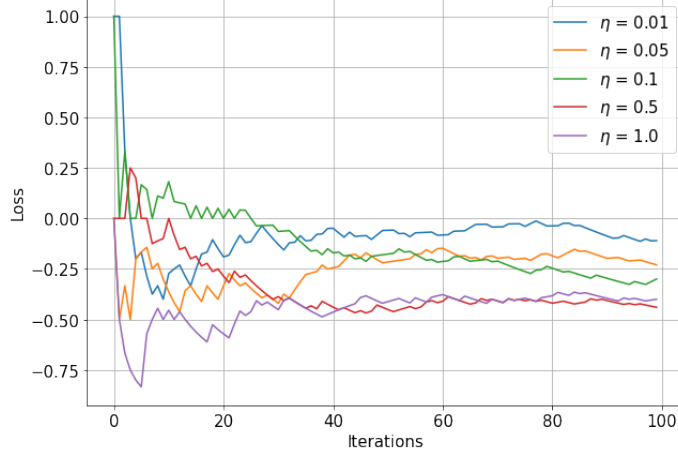


Figure 3: Evolution of average loss during 100 steps for $\eta \in 0.01, 0.05, 0.1, 0.5, 1$

According to the experimental results the best values for η seem to be 1.0 and 0.5. If we rely on our theoretical analysis, the best η should be $\sqrt{\frac{\log K}{T}}$, which for this case is $\sqrt{\frac{\log 3}{100}} \approx 0.1$. Since we are playing in a stable context, e.g the adversary is fixed so the loss doesn't change in time. Maybe using higher learning rates allow the algorithm to converge faster and act in a more 'greedy' way which works in this case because the stability of the problem.

1.3 Simulation against an adaptive adversary

- a) Consider a version of OGD with decision set $q_t = \Delta_N$ that chooses action $j_t \sim q_t$ and suffers the expected loss $l_t(q_t) = \sum_{j=1}^N q_t(j)g_t(j)$ where $g_t(j) = L(i_t, j)$. What is the update of OGD in this case? Define a function *OGDupdate* that takes as input a vector q_t , the loss vector g_t , and a learning rate $\eta > 0$ and return $q_{t+1} \in \Delta_N$

The update of OGD for the adversary j is $p_t = \Pi[p_{t-1} - \eta L(i_{t-1}, \cdot)]$. I.e the projection onto the simplex of the difference between the current weights and $\eta \times$ the loss caused by the player i choosing action i_t .

- b) Repeat the simulation of Question 2.b) when the adversary is choosing p_t with OGD with learning parameter $\eta = 0.05$

The simulation of part 2)b) but using OGD as an adversary shows that both players end up playing all actions quite uniformly.

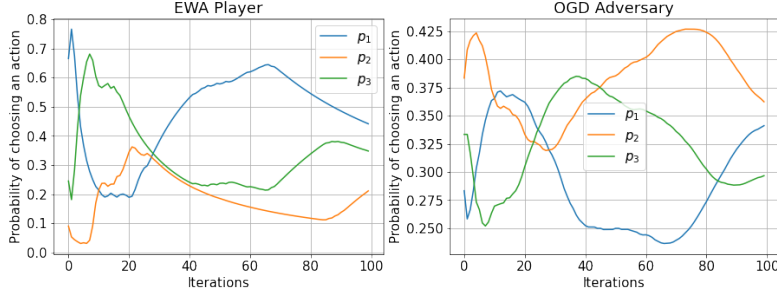


Figure 4: Evolution of the weight vectors during 100 steps for EWA vs OGD

- c) Plot $\frac{1}{t} \sum_{s=1}^t l(i_s, j_s)$ as a function of t . What strategy seems to be best between *OGD* and *EWA*?

Looking at the loss of the player (EWA) we can see that it ends up with positive loss. OGD is the winning strategy in the short run (100 steps)

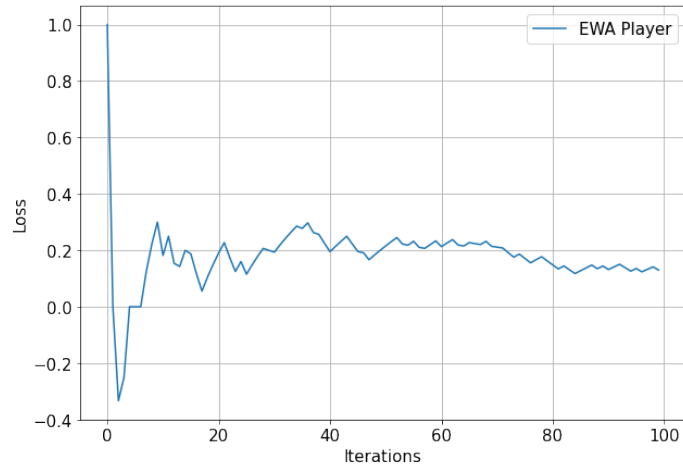


Figure 5: Evolution of average loss of EWA during 100 steps when playing against OGD

- d) Define $\hat{p}_t = \frac{1}{t} \sum_{s=1}^t p_s$. Plot in *log log scale* $\|\hat{p}_t - (1/3, 1/3, 1/3)\|_2$ as a function of $t = 1, \dots, 10000$. What do you observe?

We repeat the simulation for more steps $T = 10000$. If we keep the learning rate of EWA as $\eta = 1$, we see that after more episodes the loss approaches to 0 (reaching the equilibrium of the game). As time passes, the average weight vector comes closer to the uniform vector, however EWA doesn't converge to the uniform strategy. Note that OGD converges quickly to the uniform strategy.

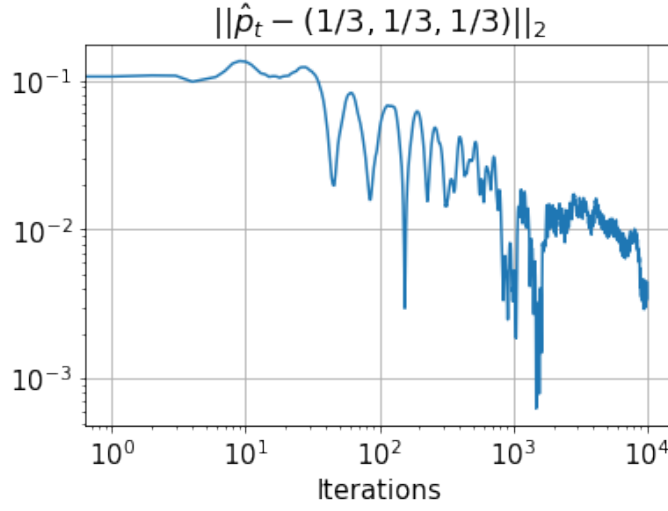


Figure 6: Evolution of the ‘distance’ to the uniform weights for EWA ($\eta = 1$) in log scale

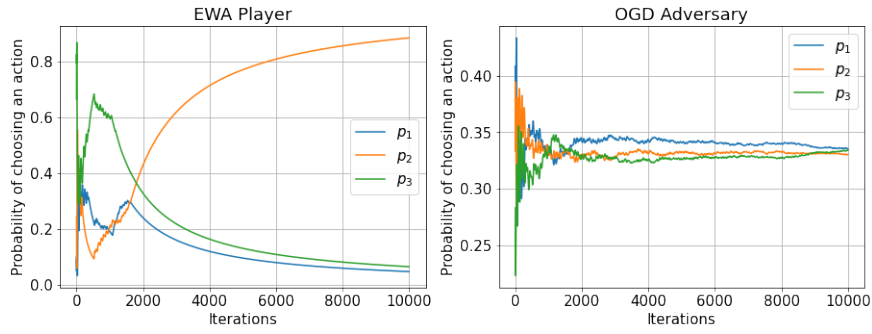


Figure 7: Evolution of the weight vectors during 10000 steps for EWA ($\eta = 1$) vs OGD

Repeating the experiment for a correctly chosen value of $\eta = 0.01$ results in having weights that are closer to the uniform both in average and in the final strategy.

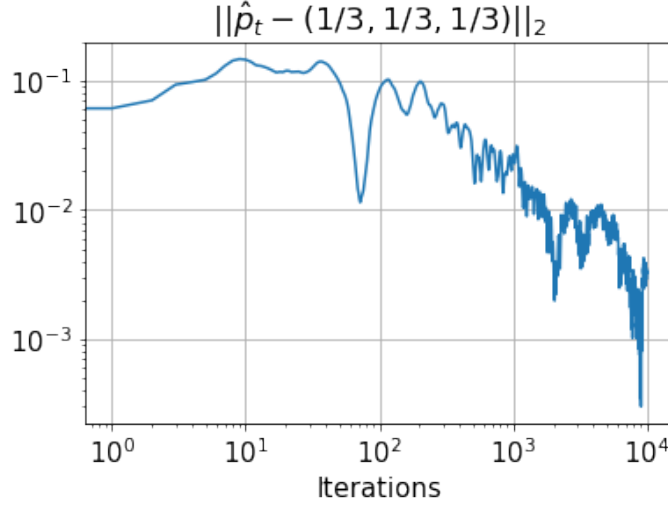


Figure 8: Evolution of the ‘distance’ to the uniform weights for EWA ($\eta = 0.01$) in log scale

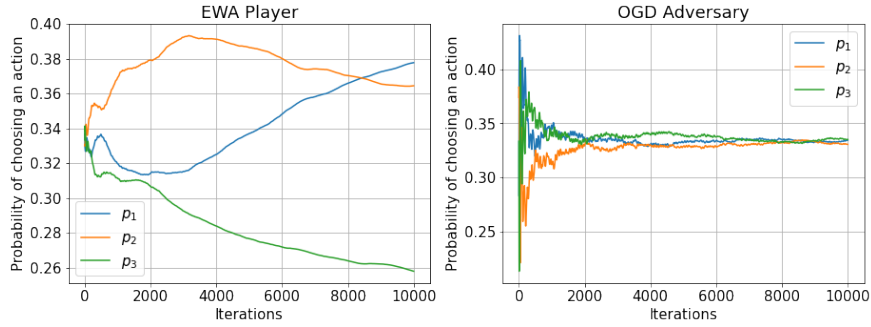


Figure 9: Evolution of the weight vectors during 10000 steps for EWA ($\eta = 0.01$) vs OGD

Bandit feedback

1.4 EXP3 vs EWA

Explain the main differences between EXP3 and EWA and implement EXP3.

EXP3 is an extension of EWA to the bandit feedback. It follows the same idea of getting an exponentially weighted average but in the case where on each round the player only observes the loss associated to the action it played. Instead of using the full loss vector, it computes an unbiased estimator of it at each round from the observed loss. Since we get less information of the environment at each step, EXP3 has a higher regret. The pseudo-regret of EXP3 has an additional $\sqrt{K}$ factor, being $R_T = 2\sqrt{KT\log K}$

1.5 EXP3 against a fixed adversary

Repeat Questions 2.b) to 2.d) with EXP3 instead of EWA.

- a) EXP3 manages to learn the same strategy as EWA, i.e playing always the action that defeats the most probable action of the adversary.

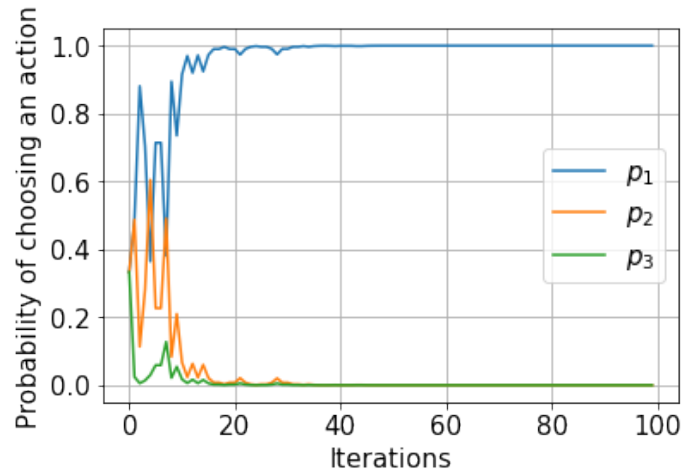


Figure 10: Evolution of the weight vectors p_i for EXP3

- b) As expected the loss remains higher than 0 for more steps than EWA, in the context of less information it takes more time to the player to learn a winning strategy. However we see that it converges to a winning strategy, ending up with a negative loss.

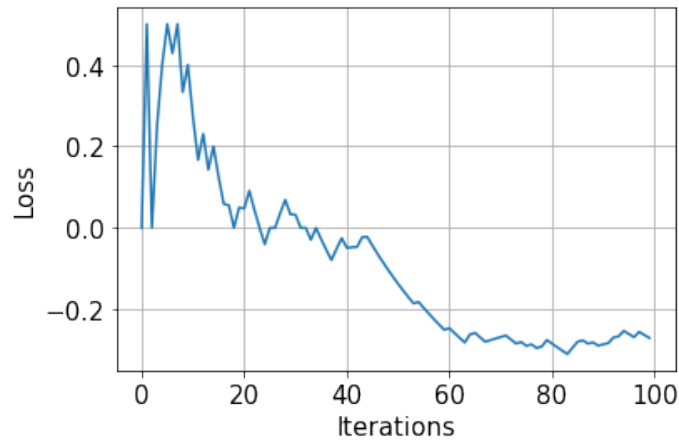


Figure 11: Evolution of average loss during 100 steps for EXP3

- c) Running the experiments for $\eta \in 0.01, 0.05, 0.1, 0.5, 1$ return the following results

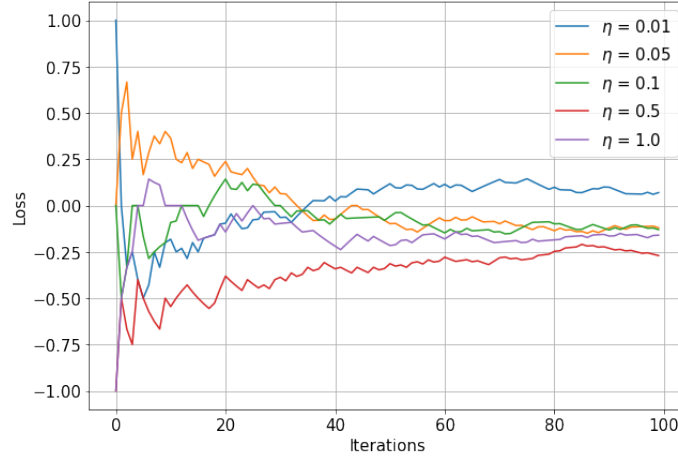


Figure 12: Evolution of average loss of EXP3 during 100 steps for $\eta \in 0.01, 0.05, 0.1, 0.5, 1$

The best theoretical value should be $\eta = \sqrt{\frac{\log K}{KT}} \approx 0.06$. Again we observe that higher learning rates work better in this ‘stable’ case, but the difference is smaller than for EWA.

Optional exercises

1.6 EXP3 vs UCB

Run a simulation where EXP3 is competing with an UCB algorithm. Who wins between UCB and EXP3? Repeat question 5) with EXP3.IX

We compare the performance between EXP3 and UCB($\sigma^2 = 1$). We select the value of η for EXP3 with the theoretical value $\sqrt{\frac{\log K}{KT}} \approx 0.02$ for 1000 iterations.

We see that after enough time, both players reach the equilibrium, although EXP3 incurs in a positive loss before converging. In this experiment, UCB seems to outperform EXP3. The weights of EXP3 oscillate during the experiment.

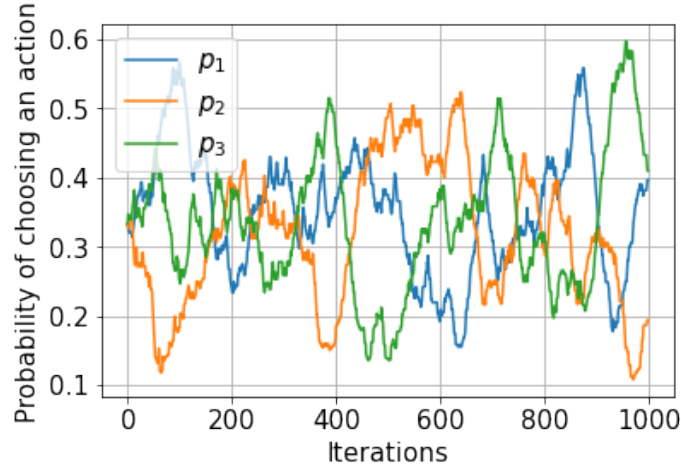


Figure 13: Evolution of the weight vectors p_i for EXP3 playing against UCB

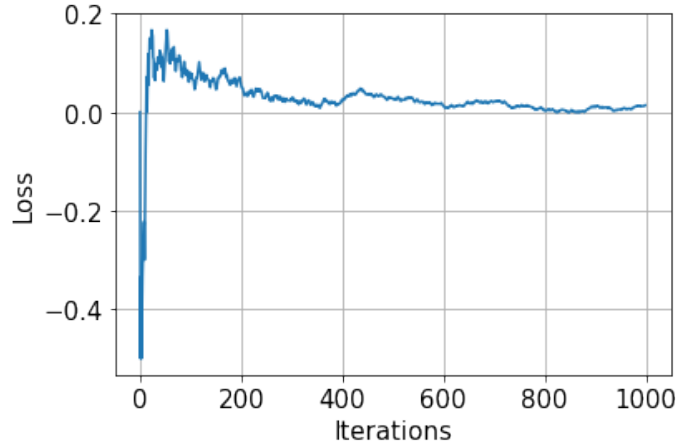


Figure 14: Evolution of average loss during 1000 steps for EXP3 playing against UCB

1.7 Prisoner's dilemma

Try different games (not necessarily zero-sum games). In particular, how these algorithms behave for the prisoner's dilemma? The prisoner's dilemma is a two-player game that shows why two completely rational individuals might not cooperate, even if it appears that is in their best interests to do so.

We compare the performance of EWA playing against OGD in the Prisoner's dilemma game. We consider the following matrices $L^{(\text{player})} = \begin{pmatrix} 1 & 3 \\ 0 & 2 \end{pmatrix}$ and

$$L^{(\text{adversary})} = \begin{pmatrix} 1 & 0 \\ 3 & 2 \end{pmatrix}$$

We observe that because each player optimizes for its own loss, both players

try to get 0 loss at the same time. This results in both of them getting a loss of 2. This is explained by the fact that for each loss vector, $(1, 0)$ or $(3, 2)$, each player minimizes ‘locally’ the loss by selecting action 2. In this case this is not enough for getting optimal performance, since this is achieved by cooperating (both players selecting action 1) and getting both a loss of 1.

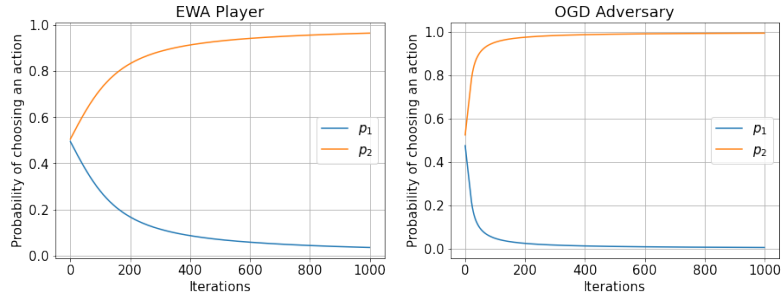


Figure 15: Evolution of the weight vectors p_i in the Prisoner's dilemma

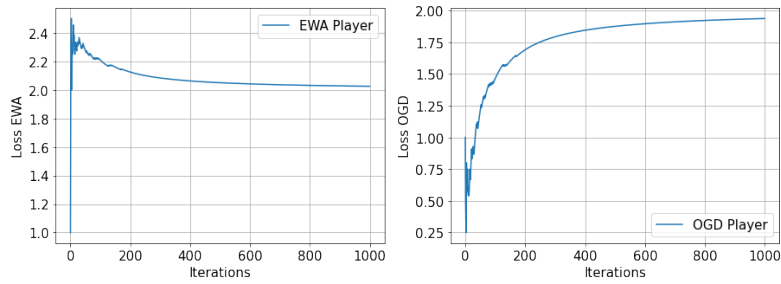


Figure 16: Evolution of average loss in the Prisoner's dilemma

2 Bernoulli Bandits

2.1 Follow the Leader

All experiments in this question will be done for $K = 2$, $p = (0.5, 0.6)$

- a) Prove that the expected regret of the Follow-the-leader algorithm (FTL) verifies $\mathbb{E}R_T \geq \alpha T$ for some $\alpha > 0$. Recall that FTL pulls at each time the arm with highest empirical mean: $k_{t+1} \arg \max_{k \in [K]} \hat{\mu}_t^k$

Let's assume that μ^1 is the arm with highest rewards (i.e $\mu^1 \geq \mu^j \forall j \in [K]$).

$$\begin{aligned} \mathbb{E}R_T &= \mathbb{E}[R_T | \hat{\mu}_1^1 = 0] \mathbb{P}(\hat{\mu}_1^1 = 0) + \mathbb{E}[R_T | \hat{\mu}_1^1 = 1] \mathbb{P}(\hat{\mu}_1^1 = 1) \\ &\geq \mathbb{E}[R_T | \hat{\mu}_1^1 = 0] \mathbb{P}(\hat{\mu}_1^1 = 0) \\ &\geq \mathbb{E}[R_T | \hat{\mu}_1^1 = 0, \hat{\mu}_1^j = 1 \text{ for some } j] \mathbb{P}(\hat{\mu}_1^1 = 0, \hat{\mu}_1^j = 1 \text{ for some } j) \\ &= T(p_1 - p_j) p_j \prod_{i \neq j} (1 - p_i) = T\alpha \end{aligned}$$

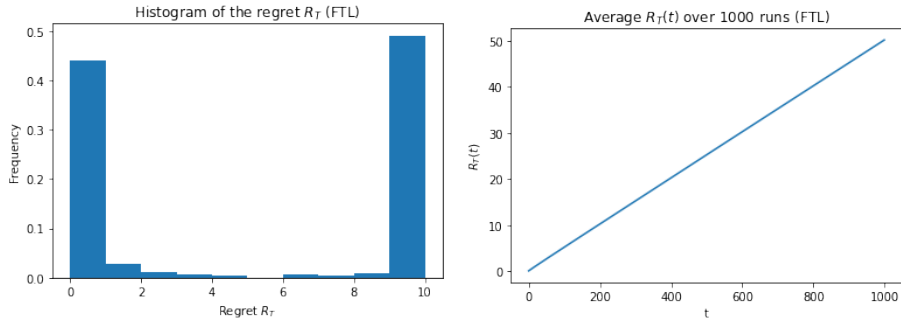
- b) Implement FTL

The following plots were generated with our implementation.

- c) For time $T = 100$, plot a histogram of the regret R_T of FTL over 1000 repetitions of the experiments. Explain the figure

We observe basically two modes on the histogram. If after the first steps of initialization we select the optimal arm, then we will always choose that arm thereafter and have zero regret. On the contrary, if we choose the sub-optimal arm after initialization then we will always choose it thereafter and incur in the maximal regret possible.

The other histogram bins correspond to cases where we get the same output for each arm a couple of times until we get different outputs and decide to continue with one arm.



(a) Histogram of the regret R_T of FTL for $T = 100$ over 1000 repetitions of the experiment (b) Mean regret R_T of FTL over 1000 repetitions as a function of time $t \in \{1, \dots, 1000\}$

Figure 17: Results of Follow The Leader (FTL) for a Bernoulli bandit with $K = 2$ arms and probabilities $\{0.5, 0.6\}$ for each arm.

- d) Plot the mean regret of FTL over 1000 repetitions as a function of $t \in 1, \dots, 1000$. Is FTL a good algorithm for stochastic bandits?

We clearly see the behavior we proved in part a), FTL has linear regret.

Follow The Leader is not a good algorithm for stochastic bandits. Recall that the initialization consists in sampling a reward for each arm for a first estimate of the means $\hat{\mu}$. The problem is that, if in this process, we sample 0 reward for the optimal arm and 1 reward for the others we will always choose a sub-optimal arm at each time step. Once we set $\hat{\mu}_* = 0$ for the optimal arm, FTL will always choose other arm.

This is particularly problematic in the case of Bernoulli bandits where the probability for each arm are very similar. The case with $K = 2$ with probabilities $\{0.5, 0.6\}$ is very illustrative. In this case the optimal arm is $i = 2$, but we can easily select the sub-optimal arm after initialization.

2.2 UCB

A random variable is said to be σ^2 -sub-Gaussian if for all $\lambda \in \mathbb{R}$, $\mathbb{E}[e^{\lambda(X - \mathbb{E}[X])}] \leq e^{\frac{1}{2}\sigma^2\lambda^2}$. The UCB(σ^2) algorithm pulls arm $k_{t+1} = \arg \min_{k \in [K]} \hat{\mu}_t^k + \sqrt{\frac{2\sigma^2 \log(t)}{N_t^k}}$. It is designed to have low regret on σ^2 -sub-Gaussian random variables.

- a) Compute the cumulant generating function, defined for $\lambda \in \mathbb{R}$ by $\phi_X(\lambda) = \log \mathbb{E}[e^{\lambda(X - \mathbb{E}[X])}]$, for a Bernoulli random variable with parameter p , denoted $\mathcal{B}p$.

We have $\mathbb{E}[X] = p$. Using the definition of expectation we get

$$\begin{aligned} \phi_X(\lambda) &= \log \mathbb{E}[e^{\lambda(X - \mathbb{E}[X])}] \\ &= \log(pe^{\lambda(1-p)} + (1-p)e^{-\lambda p}) \\ &= -\lambda p + \log(1 + p + pe^\lambda) \end{aligned}$$

- b) Prove that if a random variable X (not necessarily Bernoulli) verifies $\phi_X''(\lambda) \leq \sigma^2$ for all $\lambda \in \mathbb{R}$, then the random variable is σ^2 -sub-Gaussian

Integrating two times at both sides of the inequality $\phi_X''(\lambda) \leq \sigma^2$ we get $\phi_X(\lambda) \leq \frac{\sigma^2\lambda^2}{2}$. Then we just exponentiate $e^{\phi_X(\lambda)} \leq e^{\frac{\sigma^2\lambda^2}{2}}$, which concludes the proof since $e^{\phi_X(\lambda)} = \mathbb{E}[e^{\lambda(X - \mathbb{E}[X])}]$. This shows that X is σ -sub-Gaussian.

- c) Using question 2.b, find σ^2 such that a random variable with distribution $\mathcal{B}(p)$ is σ^2 -sub-Gaussian

We derivate two times the cumulant generating function of a Bernoulli

$$\begin{aligned} \phi_X'(\lambda) &= -p + \frac{pe^\lambda}{pe^\lambda + 1 - p} \\ \phi_X''(\lambda) &= \frac{1-p}{1-p+pe^{-\lambda}} \leq 1 \text{ for all } \lambda \end{aligned}$$

Then, using 2.b we conclude that a random variable $\mathcal{B}(p)$ is 1-sub-Gaussian.

- d) Prove that a random variable X supported on $[0, 1]$ with mean $p \in [0, 1]$ verifies $\phi_X(\lambda) \leq \phi_Y(\lambda)$ for all $\lambda \in \mathbb{R}$, where Y has a $\mathcal{B}(p)$ distribution. Hint: prove that for all $x \in [0, 1]$, for all $\lambda \in \mathbb{R}$, $e^{\lambda x} \leq 1 - x + xe^\lambda$.

Let's first prove the hint. Consider $f(x) = 1 - x + xe^\lambda - e^{\lambda x}$. $f(0) = f(1) = 0$. We can show that the function is concave downwards, $f''(x) = -\lambda^2 e^{\lambda x} \leq 0$ for all λ and x . Then, $f(x) \geq 0$ for all $x \in [0, 1]$ because $f(0) = f(1) = 0$ and the function is concave downwards.

$$\begin{aligned}\phi_X(\lambda) &= \log \mathbb{E}[e^{\lambda(X-p)}] \\ &= -\lambda p + \log \mathbb{E}[e^{\lambda X}] \\ &\leq -\lambda p + \log \mathbb{E}[1 - X + Xe^\lambda] \\ &= -\lambda p + \log(1 - p + pe^\lambda) \\ &= \phi_Y(\lambda)\end{aligned}$$

- e) Prove that all random variables supported on $[0, 1]$ are sub-Gaussian.

From the previous part we know that for a random variable X bounded in $[0, 1]$, we have $\phi_X(\lambda) \leq \phi_Y(\lambda)$ where Y has a $\mathcal{B}(p)$ distribution. We already proved that Bernoulli random variables are 1-sub-Gaussian, so $\phi_Y(\lambda) \leq \frac{\lambda^2}{2}$. Therefore $\phi_X(\lambda) \leq \frac{\lambda^2}{2}$ and X is 1-sub-Gaussian.

- f) Implement the UCB(σ^2) algorithm.

The following plots were generated with our implementation.

- g) Plot the mean regret of UCB($\frac{1}{4}$) as a function of time up to $T = 1000$ for $K = 2$, $p = (0.5, 0.6)$, over 1000 repetitions. Compare with the result of question 1.d

We compare the performance of UCB and FTL for a Bernoulli bandit with $K = 2$ arms and probabilities $\{0.5, 0.6\}$. Unlike FTL, UCB achieves sub-linear regret, as shown in Figure 18

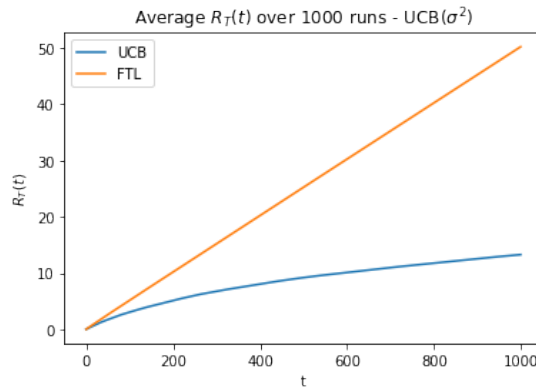


Figure 18: Comparison between FTL and UCB with $\sigma^2 = \frac{1}{4}$. UCB achieves sub-linear regret.

- h) For $K = 2$, $p = (0.6, 0.5)$, $T = 1000$, plot the mean regret of $UCB(\sigma^2)$ over 1000 repetitions as a function of σ^2 , for $\sigma^2 \in 0, 1/32, 1/16, 1/4, 1$. Do it again for $p = (0.85, 0.95)$ and compare the results: does the optimal parameter change? How does it compare to the theoretic parameter?

In the case $p = \{0.5, 0.6\}$ the best performance is achieved with $\sigma^2 = \frac{1}{4}$. Note however that this regret is very close to the one obtained with $\sigma^2 = \frac{1}{16}$. Here the value matches the one from the theoretic analysis. In both cases $p \simeq \frac{1}{2}$ and for that probability a Bernoulli variable can be shown to be $\frac{1}{4}$ -sub-Gaussian.

Moreover, we can show that for a $\mathcal{B}(p)$ we get tight sub-Gaussian constants depending on p

$$\sigma^2(p) = \begin{cases} 0 & \text{if } p \in \{0, 1\} \\ \frac{1}{4} & \text{if } p = \frac{1}{2} \\ \frac{1}{2} \frac{p - (1-p)}{\log p - \log(1-p)} & \text{otherwise} \end{cases}$$

For the second case is $p = \{0.85, 0.95\}$. For probability $p \simeq 0.9$ we have the theoretic value of $\sigma^2 \simeq 0.18$. However, in this case the best empirical results are achieved with the sub-Gaussian constant $\sigma^2 = \frac{1}{16} = 0.0625$. This might be explained by looking at the plot for next question. For values of p close to 1, the bound between σ^2 and the variance of the Bernoulli variable gets looser. It may be the case that in practice we're allowed to consider smaller confidence intervals that cause faster convergence but still retain high probability guarantees. If this is the case, using smaller values for σ^2 might cause UCB to converge faster to the optimal arm, therefore we get a lower regret.

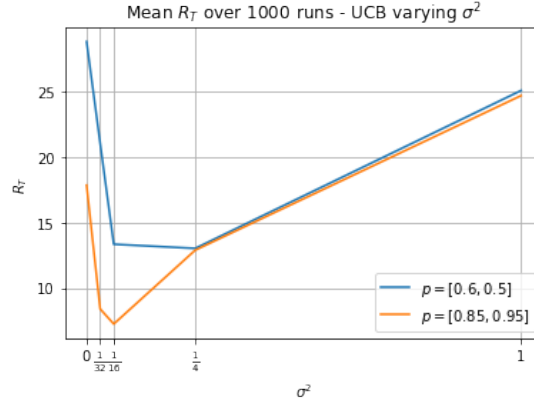


Figure 19: UCB results varying the σ^2 sub-Gaussian constant.

2.3

On the same figure, plot the variance of $\mathcal{B}(p)$ and the sub-Gaussian constant $\sigma^2(p)$ described above as a function of $p \in [0, 1]$.

Finally, we observe that for a $\mathcal{B}(p)$, the variance is bounded by the $\sigma^2(p)$, see Figure 20.

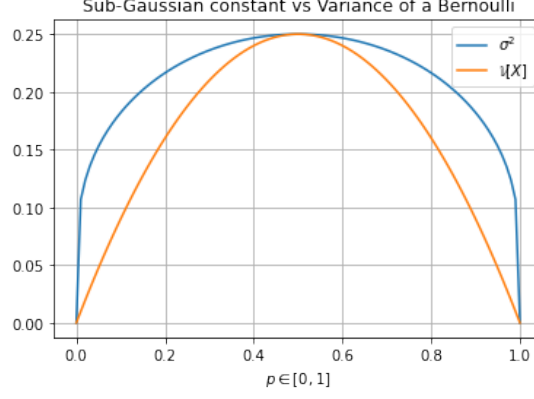


Figure 20: Variance of a Bernoulli and $\sigma^2(p)$.

2.4

(optional) Prove that a σ^2 -sub-Gaussian random variable has variance bounded by σ^2 .

Adaptation to the variance.

2.5 UCB-V

a) *Prove that $N_t^k \hat{v}_t^k = \sum_{s=1}^t 1\{k_s = k\} (X_s^{k_s})^2 - \frac{1}{N_t^k} \left(\sum_{s=1}^t 1\{k_s = k\} X_s^{k_s} \right)^2$*

$$\begin{aligned} N_t^k \hat{v}_t^k &= \sum_{s=1}^t 1\{k_s = k\} (X_s^{k_s})^2 + \sum_{s=1}^t 1\{k_s = k\} (\hat{\mu}_t^k)^2 - 2 \sum_{s=1}^t 1\{k_s = k\} X_s^{k_s} \hat{\mu}_t^k \\ &= \sum_{s=1}^t 1\{k_s = k\} (X_s^{k_s})^2 + N_t^k (\hat{\mu}_t^k)^2 - 2 N_t^k (\hat{\mu}_t^k)^2 \\ &= \sum_{s=1}^t 1\{k_s = k\} (X_s^{k_s})^2 - \frac{1}{N_t^k} \left(\sum_{s=1}^t 1\{k_s = k\} X_s^{k_s} \right)^2 \end{aligned}$$

b) *Prove that $N_{t+1}^{k+1} \hat{v}_{t+1}^{k+1} = N_t^{k+1} \hat{v}_t^{k+1} + (X_{t+1}^{k_{t+1}} - \hat{\mu}_t^{k_{t+1}})(X_{t+1}^{k_{t+1}} - \hat{\mu}_{t+1}^{k_{t+1}})$. What is the practical advantage of that formulation?*

$$\begin{aligned} N_{t+1}^{k+1} \hat{v}_{t+1}^{k+1} &= \sum_{s=1}^{t+1} 1\{k_s = k_{t+1}\} (X_s^{k_s})^2 - N_{t+1}^{k+1} (\hat{\mu}_{t+1}^{k_{t+1}})^2 \\ &= \sum_{s=1}^t 1\{k_s = k_{t+1}\} (X_s^{k_s})^2 - N_t^{k+1} (\hat{\mu}_t^{k_{t+1}})^2 + (X_{t+1}^{k_{t+1}})^2 + N_t^{k+1} (\hat{\mu}_t^{k_{t+1}})^2 - N_{t+1}^{k+1} (\hat{\mu}_{t+1}^{k_{t+1}})^2 \\ &= N_t^{k+1} \hat{v}_t^{k+1} + (X_{t+1}^{k_{t+1}})^2 + N_t^{k+1} (\hat{\mu}_t^{k_{t+1}})^2 - N_{t+1}^{k+1} (\hat{\mu}_{t+1}^{k_{t+1}})^2 \end{aligned}$$

Since $N_{t+1}^{k_{t+1}} \hat{\mu}_{t+1}^{k_{t+1}} = N_t^{k_{t+1}} \hat{\mu}_t^{k_{t+1}} + X_{t+1}^{k_{t+1}}$ and $N_{t+1}^{k_{t+1}} = N_t^{k_{t+1}} + 1$, we can write the last part of the equation as

$$\begin{aligned} N_t^{k_{t+1}} (\hat{\mu}_t^{k_{t+1}})^2 - N_{t+1}^{k_{t+1}} (\hat{\mu}_{t+1}^{k_{t+1}})^2 &= (N_{t+1}^{k_{t+1}} \hat{\mu}_{t+1}^{k_{t+1}} - X_{t+1}^{k_{t+1}}) \hat{\mu}_t^{k_{t+1}} - N_t^{k_{t+1}} \hat{\mu}_t^{k_{t+1}} \hat{\mu}_{t+1}^{k_{t+1}} - X_{t+1}^{k_{t+1}} \hat{\mu}_{t+1}^{k_{t+1}} \\ &= \hat{\mu}_{t+1}^{k_{t+1}} \hat{\mu}_t^{k_{t+1}} - X_{t+1}^{k_{t+1}} \hat{\mu}_t^{k_{t+1}} - X_{t+1}^{k_{t+1}} \hat{\mu}_{t+1}^{k_{t+1}} \end{aligned}$$

We conclude the proof by grouping the terms

$$(X_{t+1}^{k_{t+1}})^2 + \hat{\mu}_{t+1}^{k_{t+1}} \hat{\mu}_t^{k_{t+1}} - X_{t+1}^{k_{t+1}} \hat{\mu}_t^{k_{t+1}} - X_{t+1}^{k_{t+1}} \hat{\mu}_{t+1}^{k_{t+1}} = (X_{t+1}^{k_{t+1}} - \hat{\mu}_t^{k_{t+1}})(X_{t+1}^{k_{t+1}} - \hat{\mu}_{t+1}^{k_{t+1}})$$

The advantage of this formulation is that it allow us to compute the variance estimate with a recursive update (just as we do with the average). This has two advantages with respect to the classic formula: i) we don't need to keep record of all the samples (advantage in space) and ii) is more stable.

c) *Implement UCB-V*

The following plots were generated with our implementation.

d) *On the same figure, plot the mean regret of UCB-V and UCB(1/4) as a function of time up to $T = 1000$ for $K = 2$, $p = (0.5, 0.6)$, over 1000 repetitions*

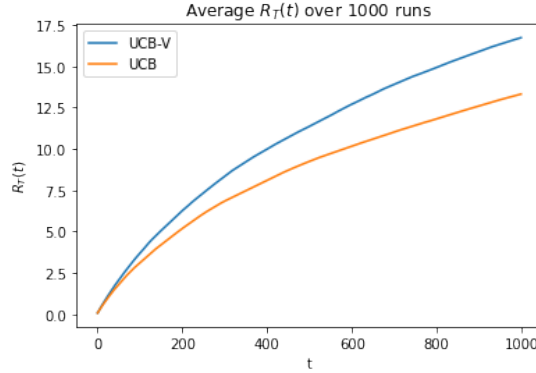


Figure 21: Mean regret R_T over 1000 repetitions for UCB-V and UCB with $\sigma^2 = \frac{1}{4}$ for a Bernoulli bandit with $K = 2$ arms and $p = \{0.5, 0.6\}$

e) *Same question for $p = (0.1, 0.2)$ and $p = (0, 0.1)$. Compare to the results of 5.d. When does UCB-V improve over UCB?*

In general UCB-V improves over UCB when we don't know in advance a tight sub-Gaussian constant. In the examples, we use the highest possible sub-Gaussian constant $\sigma^2 = \frac{1}{4}$ for all experiments. In the case $p = \{0.5, 0.6\}$ UCB is actually better than UCB-V because $\sigma^2 = \frac{1}{4}$ is tight. The variance estimation of UCB-V can't compete the theoretical

correct value of σ^2 used in UCB. However, for the cases $p = \{0.1, 0.2\}$ and $p = \{0, 0.1\}$ UCB-V achieves better results. The difference is particularly big in the case $p = \{0, 0.1\}$. For this probabilities the sub-Gaussian constant should be very small for $p = 0$ ($\sigma^2 = 0$) instead of $\sigma^2 = \frac{1}{4}$. Unlike UCB, UCB-V can adapt its estimate $\hat{v}_t^k$ to be small and produce tighter confidence intervals, thus obtaining better performance.

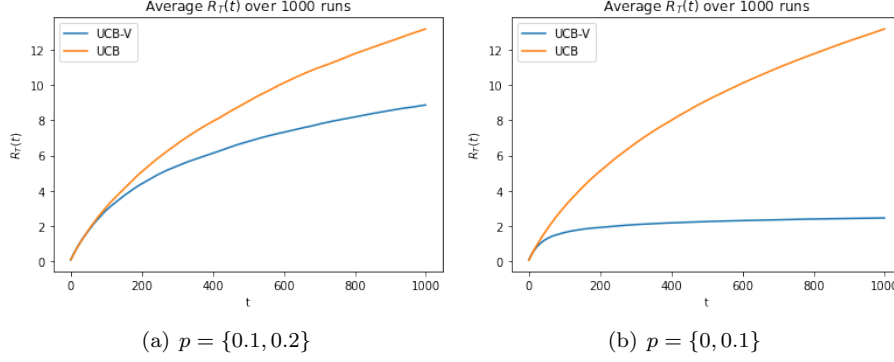


Figure 22: Mean regret R_T over 1000 repetitions for UCB-V and UCB with $\sigma^2 = \frac{1}{4}$ for two Bernoulli bandits with $K = 2$ arms